Orbital Box Worksheet

ANSWER KEY · TEACHER COPY

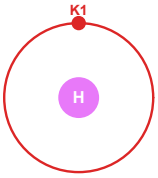
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

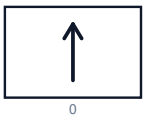
Z = 1 H · Hydrogen

period 1 · group 1 · s-block · 1 valence electrons

Bohr model



1s



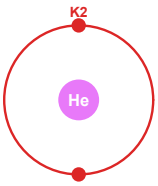
Shells: 1 · K 1 · valence 1
Energy order: $1s^1$
Configuration: $1s^1$

cut here

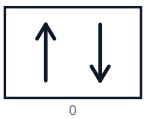
Z = 2 He · Helium

period 1 · group 18 · s-block · 2 valence electrons

Bohr model



1s



Shells: 2 · K 2 · valence 2
Energy order: $1s^2$
Configuration: $1s^2$

Orbital Box Worksheet

ANSWER KEY · TEACHER COPY

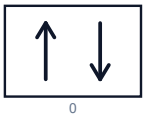
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 3 Li · Lithium

period 2 · group 1 · s-block · 1 valence electrons

1s



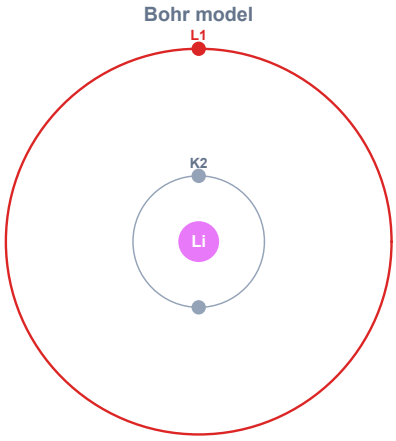
0

2s



0

Bohr model



Shells: 2, 1 · K 2 · L 1 · valence 1

Energy order: $1s^2\ 2s^1$

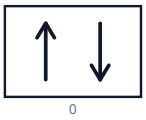
Configuration: $1s^2\ 2s^1$

cut here

Z = 4 Be · Beryllium

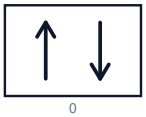
period 2 · group 2 · s-block · 2 valence electrons

1s



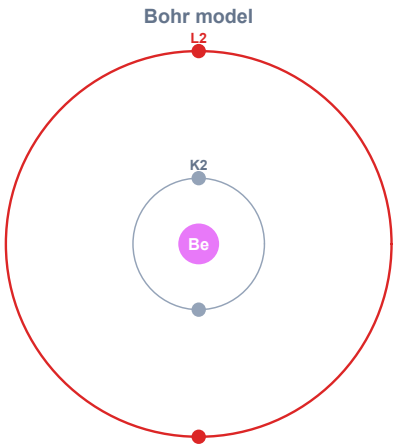
0

2s



0

Bohr model



Shells: 2, 2 · K 2 · L 2 · valence 2

Energy order: $1s^2\ 2s^2$

Configuration: $1s^2\ 2s^2$

Orbital Box Worksheet

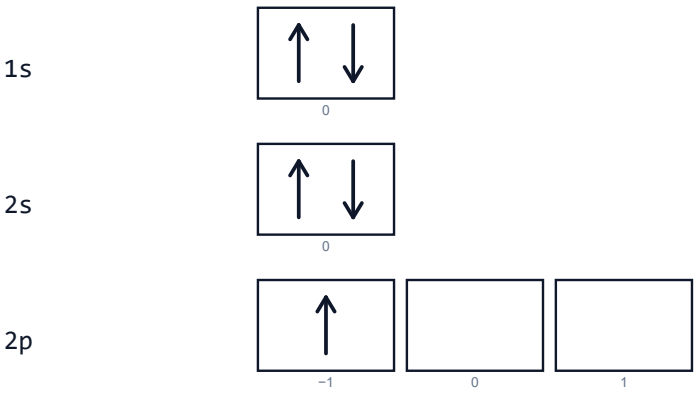
ANSWER KEY · TEACHER COPY

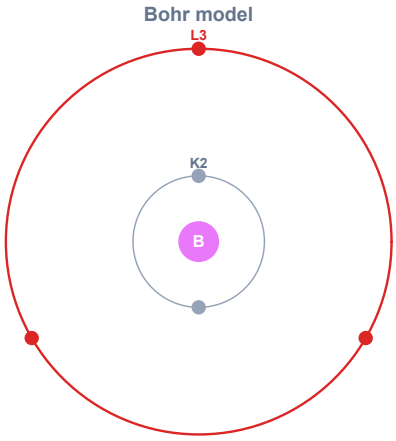
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 5 B · Boron

period 2 · group 13 · p-block · 3 valence electrons





Shells: 2, 3 · K 2 · L 3 · valence 3

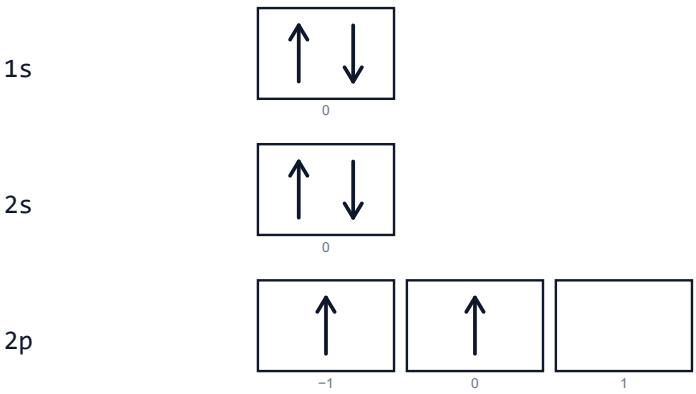
Energy order: $1s^2\ 2s^2\ 2p^1$

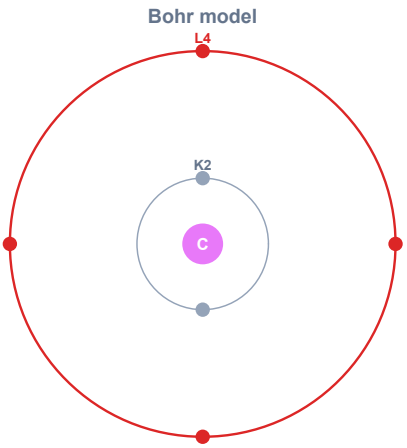
Configuration: $1s^2\ 2s^2\ 2p^1$

cut here

Z = 6 C · Carbon

period 2 · group 14 · p-block · 4 valence electrons





Shells: 2, 4 · K 2 · L 4 · valence 4

Energy order: $1s^2\ 2s^2\ 2p^2$

Configuration: $1s^2\ 2s^2\ 2p^2$

Orbital Box Worksheet

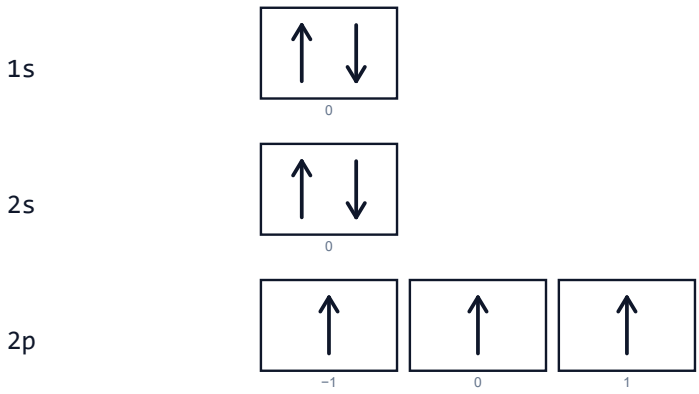
ANSWER KEY · TEACHER COPY

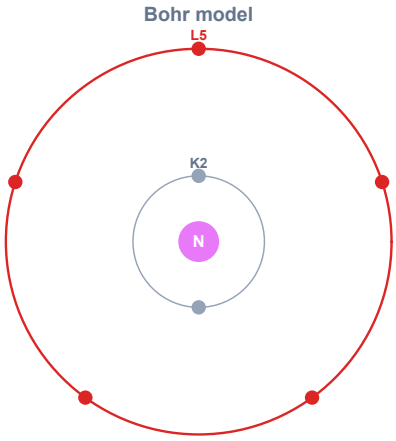
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 7 N · Nitrogen

period 2 · group 15 · p-block · 5 valence electrons





Shells: 2, 5 · K 2 · L 5 · valence 5

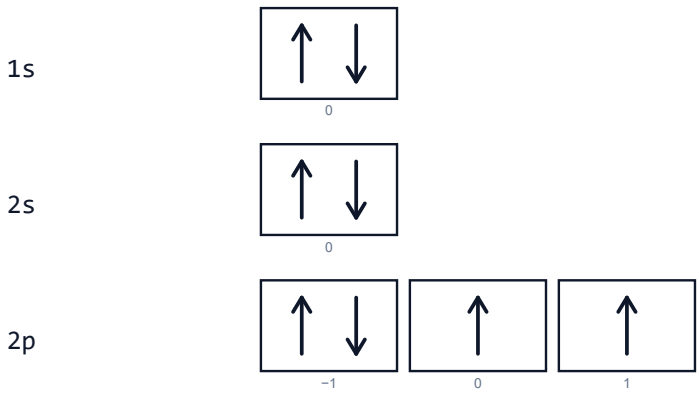
Energy order: $1s^2\ 2s^2\ 2p^3$

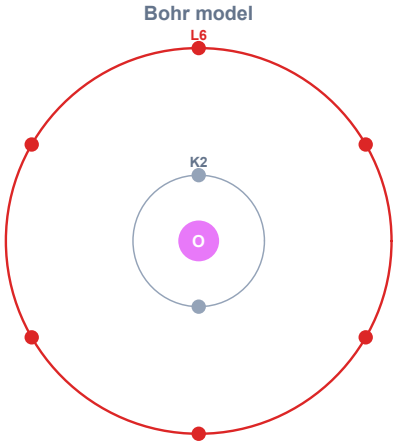
Configuration: $1s^2\ 2s^2\ 2p^3$

cut here

Z = 8 O · Oxygen

period 2 · group 16 · p-block · 6 valence electrons





Shells: 2, 6 · K 2 · L 6 · valence 6

Energy order: $1s^2\ 2s^2\ 2p^4$

Configuration: $1s^2\ 2s^2\ 2p^4$

Orbital Box Worksheet

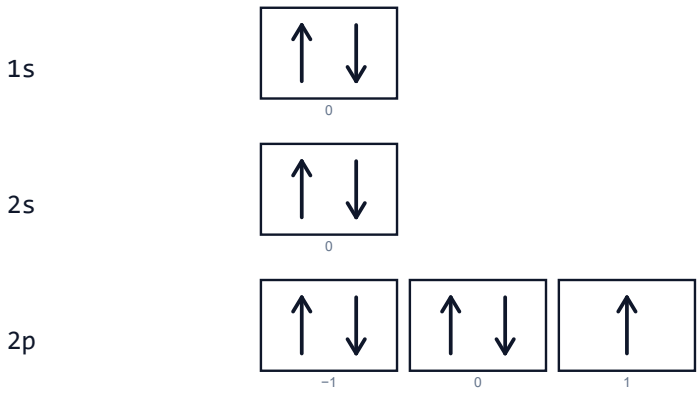
ANSWER KEY · TEACHER COPY

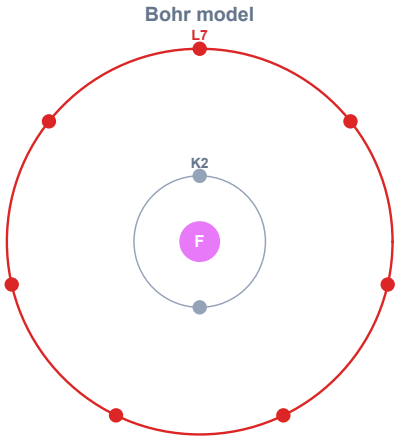
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 9 F · Fluorine

period 2 · group 17 · p-block · 7 valence electrons





Shells: 2, 7 · K 2 · L 7 · valence 7

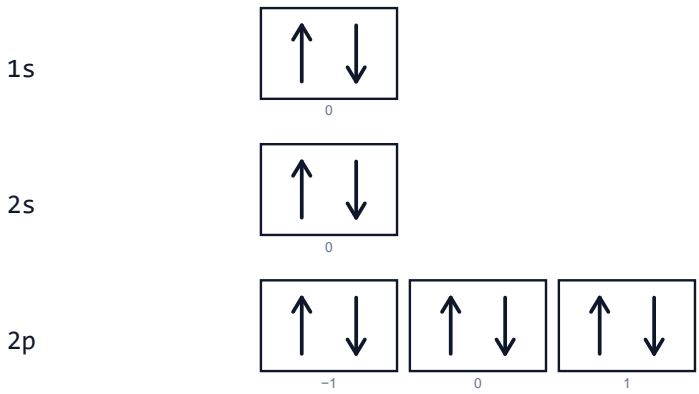
Energy order: $1s^2\ 2s^2\ 2p^5$

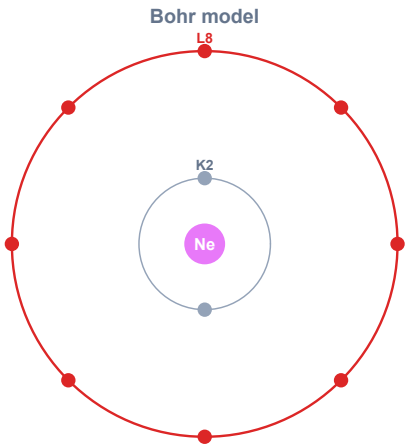
Configuration: $1s^2\ 2s^2\ 2p^5$

cut here

Z = 10 Ne · Neon

period 2 · group 18 · p-block · 8 valence electrons





Shells: 2, 8 · K 2 · L 8 · valence 8

Energy order: $1s^2\ 2s^2\ 2p^6$

Configuration: $1s^2\ 2s^2\ 2p^6$

Orbital Box Worksheet

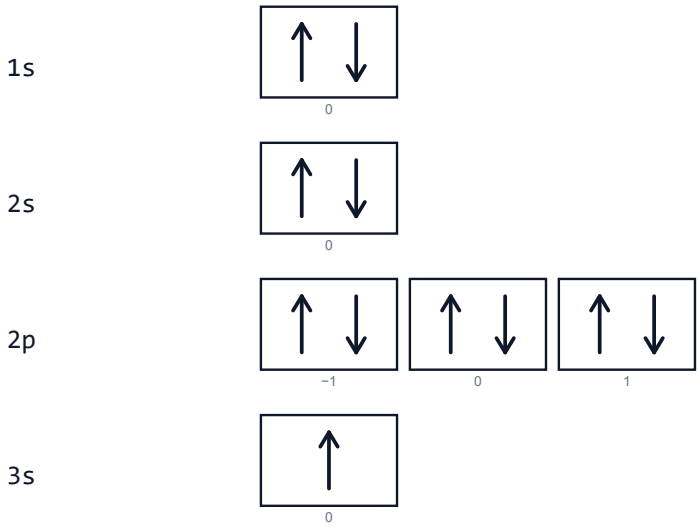
ANSWER KEY · TEACHER COPY

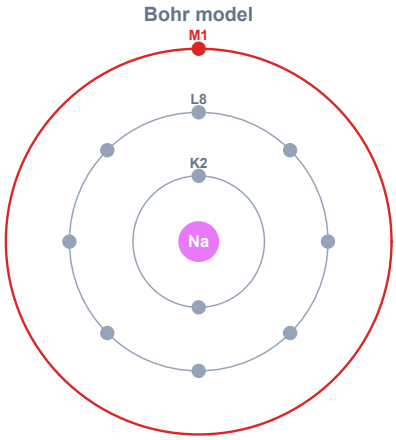
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 11 Na · Sodium

period 3 · group 1 · s-block · 1 valence electrons





Shells: 2, 8, 1 · K 2 · L 8 · M 1 · valence 1

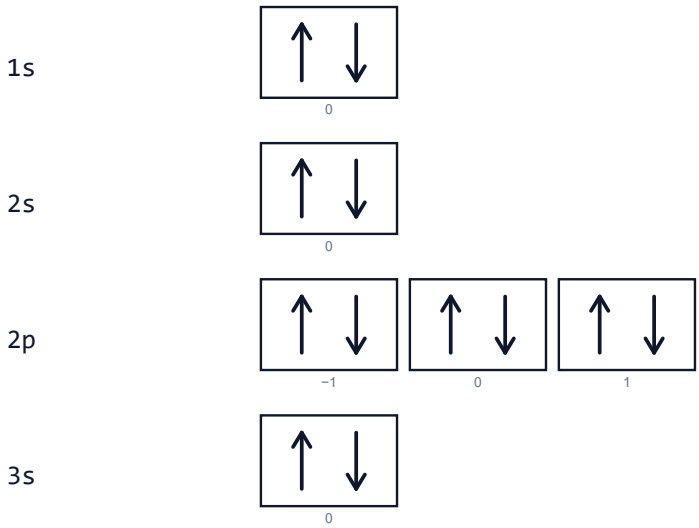
Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^1$

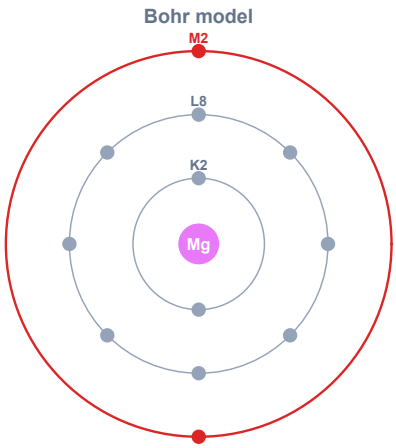
Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^1$

cut here

Z = 12 Mg · Magnesium

period 3 · group 2 · s-block · 2 valence electrons





Shells: 2, 8, 2 · K 2 · L 8 · M 2 · valence 2

Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2$

Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2$

Orbital Box Worksheet

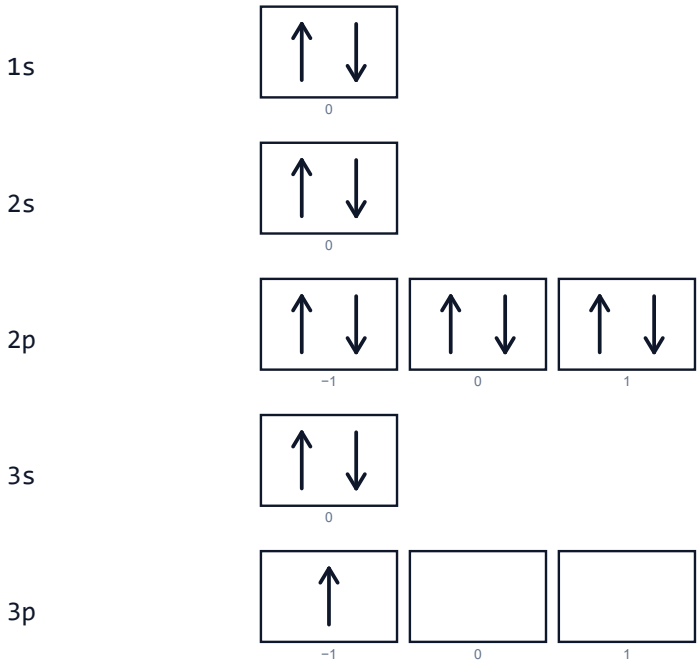
ANSWER KEY · TEACHER COPY

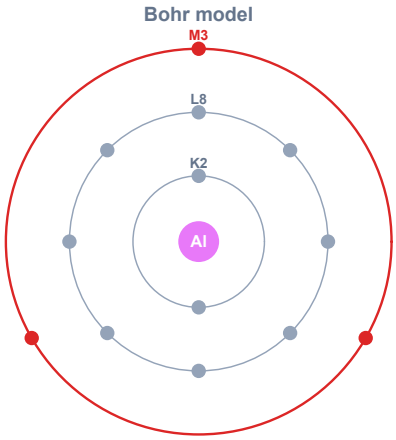
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 13 Al · Aluminium

period 3 · group 13 · p-block · 3 valence electrons





Shells: 2, 8, 3 · K 2 · L 8 · M 3 · valence 3

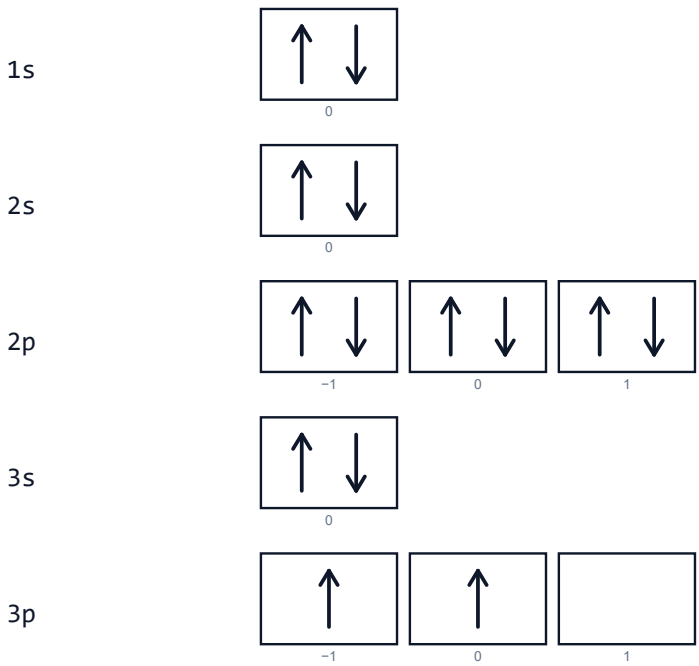
Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

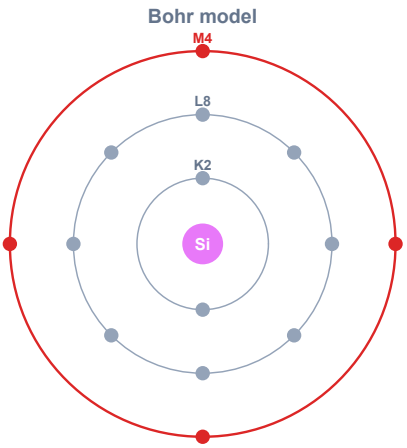
Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

cut here

Z = 14 Si · Silicon

period 3 · group 14 · p-block · 4 valence electrons





Shells: 2, 8, 4 · K 2 · L 8 · M 4 · valence 4

Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

Orbital Box Worksheet

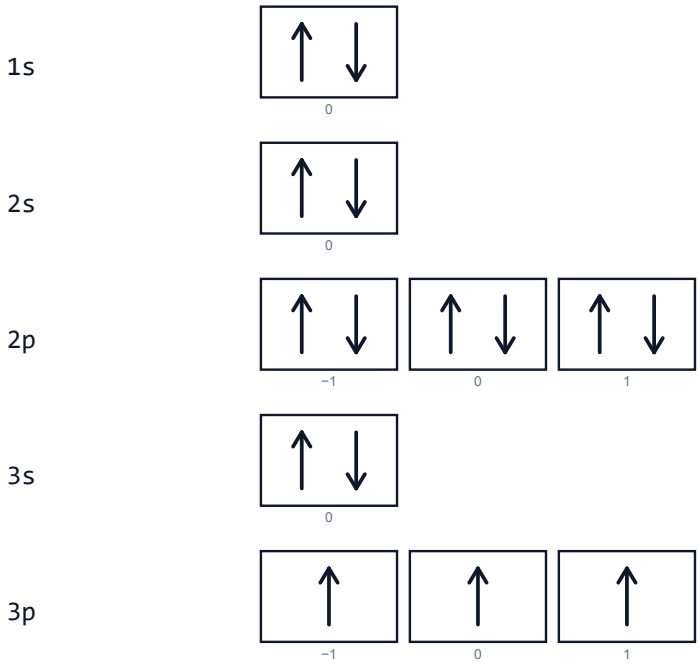
ANSWER KEY · TEACHER COPY

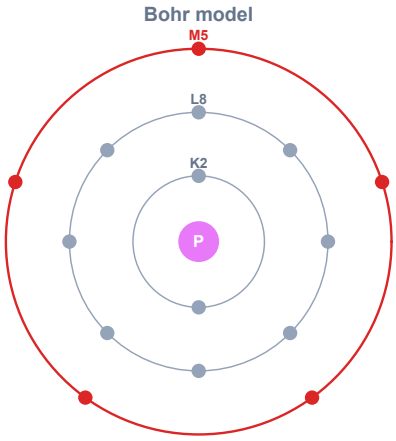
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 15 P · Phosphorus

period 3 · group 15 · p-block · 5 valence electrons





Shells: 2, 8, 5 · K 2 · L 8 · M 5 · valence 5

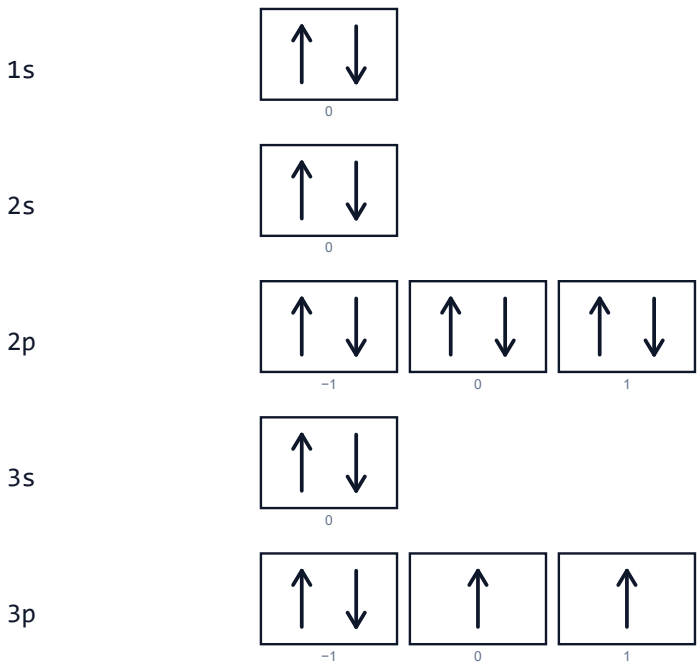
Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

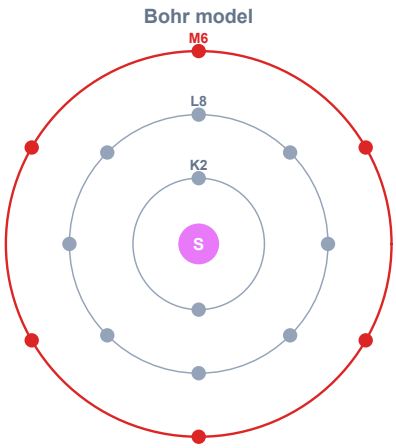
Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

cut here

Z = 16 S · Sulfur

period 3 · group 16 · p-block · 6 valence electrons





Shells: 2, 8, 6 · K 2 · L 8 · M 6 · valence 6

Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$

Orbital Box Worksheet

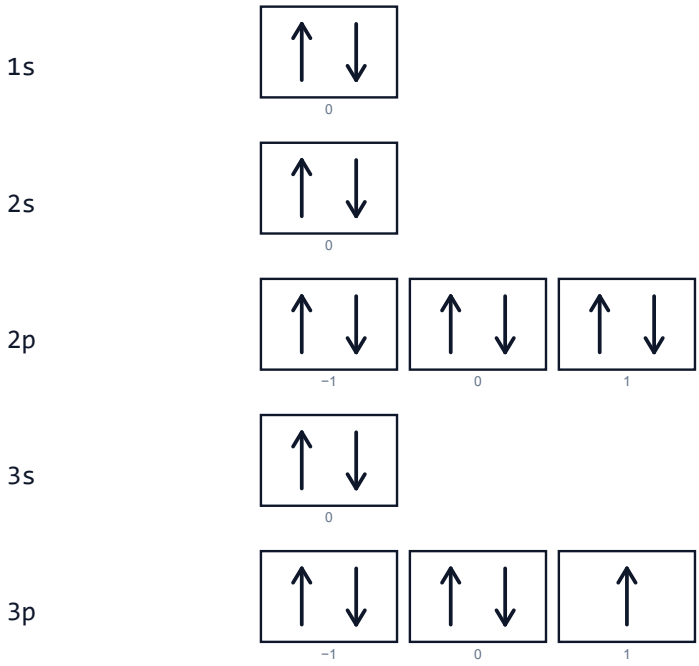
ANSWER KEY · TEACHER COPY

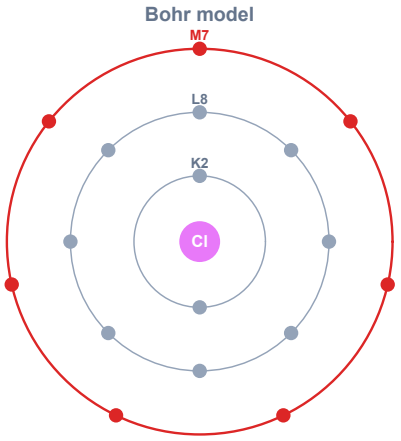
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 17 Cl · Chlorine

period 3 · group 17 · p-block · 7 valence electrons





Shells: 2, 8, 7 · K 2 · L 8 · M 7 · valence 7

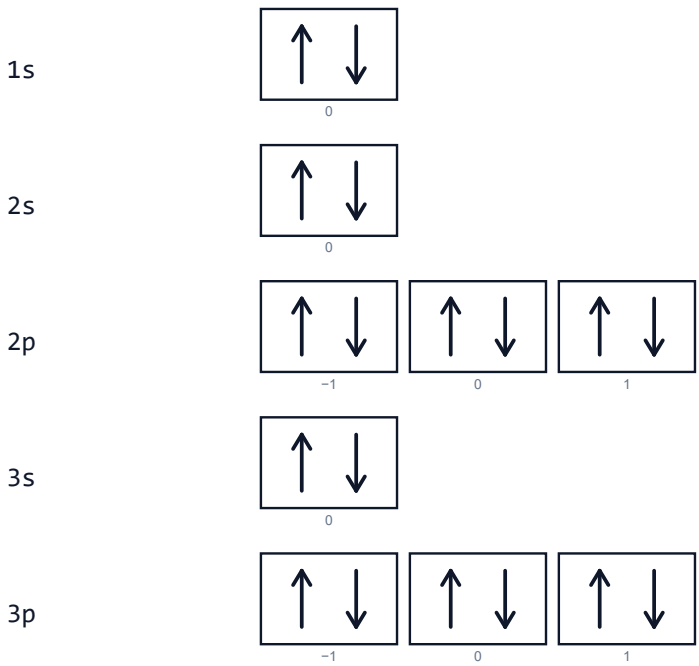
Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

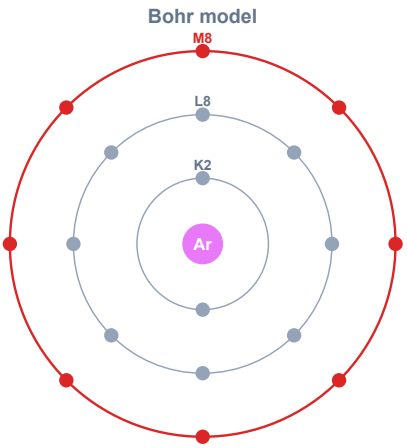
Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

cut here

Z = 18 Ar · Argon

period 3 · group 18 · p-block · 8 valence electrons





Shells: 2, 8, 8 · K 2 · L 8 · M 8 · valence 8

Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Orbital Box Worksheet

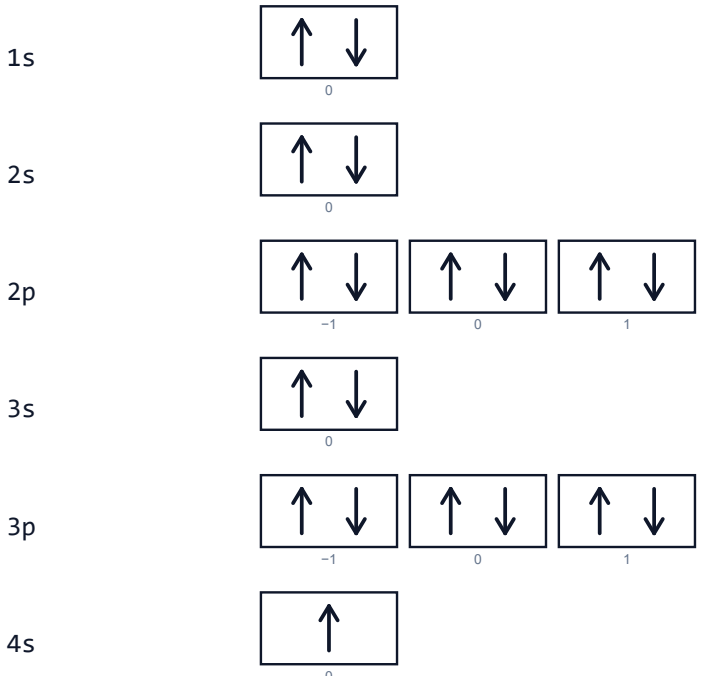
ANSWER KEY · TEACHER COPY

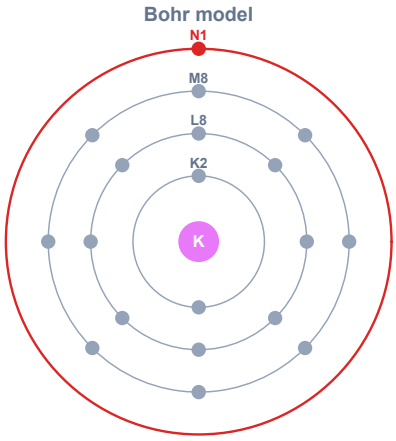
Teacher: _____ Class: _____ Section: _____ Date: November 07, 2026

Teacher copy: the boxes carry the correct spins; the configuration, energy order and shell counts are printed under each diagram.

Z = 19 K · Potassium

period 4 · group 1 · s-block · 1 valence electrons





Shells: 2, 8, 8, 1 · K 2 · L 8 · M 8 · N 1 · valence 1

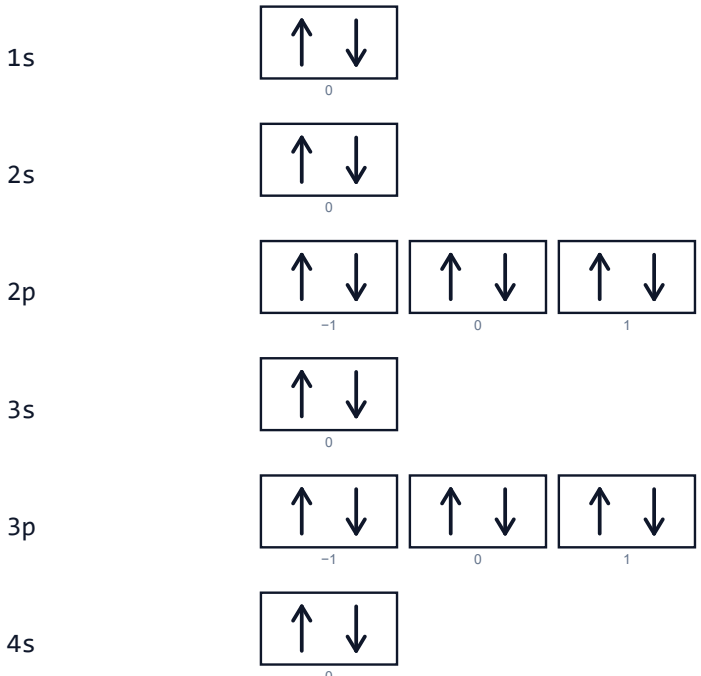
Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

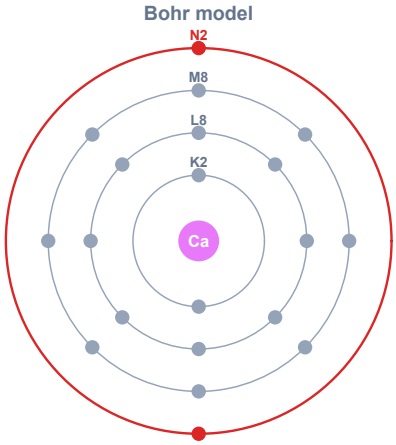
Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

cut here

Z = 20 Ca · Calcium

period 4 · group 2 · s-block · 2 valence electrons





Shells: 2, 8, 8, 2 · K 2 · L 8 · M 8 · N 2 · valence 2

Energy order: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

Configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$